QISKIT FALL FEST 2026 · ADVANCED TRACK · ERROR DETECTION · [Team Name: Quantum Eclipse]

Spacetime Sentinels

A game of low-overhead quantum error detection
with spacetime codes. Every score is computed with Qiskit.

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Live game

arXiv quant-ph · submitted

Scan to play

THE QUESTION

Can you protect a quantum computer without fixing a single error?

Our track: **error detection with spacetime codes** (`qiskit-paulice`). The rule we set ourselves: **the rules are the physics, and every score comes from Qiskit.**

Detection

THIS GAME

Helper qubits flag the errors they can see. Those shots are thrown away. Nothing is repaired.

Mitigation

Fix the *estimate* afterwards: ZNE, PEC, twirling, readout correction.

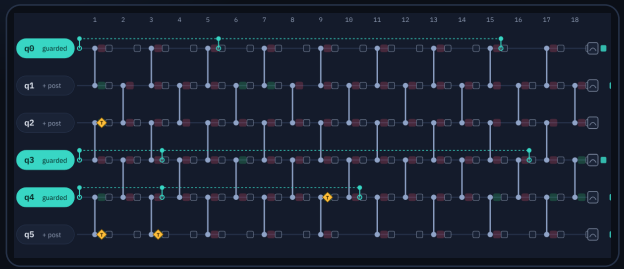
Correction

Logical qubits, repeated stabilizers and a decoder that undoes errors mid-run.

Detection is the cheapest of the three. The game asks: **how should you spend it?**

THE PHYSICS

A spacetime check in one picture



Live board, depth 18. Teal lines are three checks; gold diamonds are T gates on Z-safe sites.

- One **ancilla** watches one qubit.
- It couples with a controlled X, Y or Z at a few **moments in time**.
- Syndrome = ancilla bit $\oplus$ parity of some outputs. Always 0 if nothing went wrong.

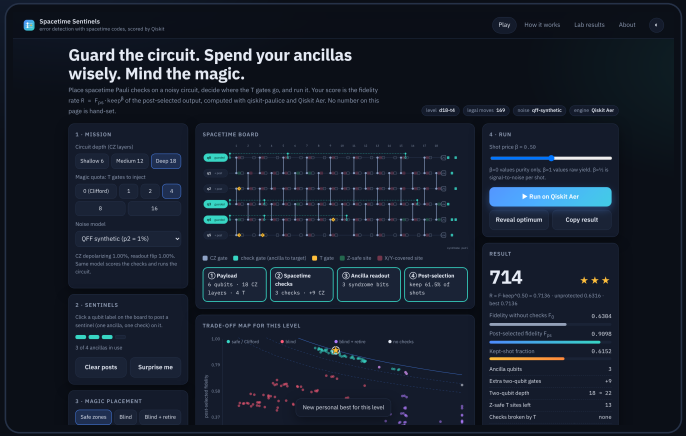
The catch: T gates

$TZT^\dagger = Z$ but $TXT^\dagger = \frac{X+Y}{\sqrt{2}}$

A T where a check acts as X/Y **breaks** it. On IBM hardware T is a virtual R_Z: **free in noise, not free for the code.**

PLAY

The game: four decisions, one score



1. **Mission.** Depth, *T* quota, noise.
2. **Sentinels.** Post up to 4 checks.
3. **Magic.** Safe zones, blind, or blind + retire.
4. **Run.** Qiskit Aer scores the move.

Scan to play

DESIGN

Every rule is a physics decision

GAME RULE	PHYSICS	QISKIT
Post a sentinel	one ancilla + one check, budget of 4	`add_pauli_checks`
Green / red sites	every check acts as I/Z there, or not	`uncovered_paulis`
Magic quota	non-Clifford T gates added after the checks	`QuantumCircuit`
Safe / Blind / Retire	doping policy; broken checks found exactly	`Statevector`
Run	payload $\rightarrow$ checks $\rightarrow$ readout $\rightarrow$ post-select	Aer density matrix
Score	$1000\, F_{ps}\kappa^{\beta}$, with $\beta=\frac{1}{2}$ the SNR per $\sqrt{\text{shot}}$	`sentinels.scoring`

5,412 legal moves | **all evaluated exactly in Qiskit Aer** | **no hand-set numbers**

SCORING

Why the score is $R = F_{\mathrm{ps}}\sqrt{\kappa}$

$$R = F_{\mathrm{ps}}\,\kappa^{1/2}$$

Shots needed $\propto$ R^{-2}

quality × quantity

- Residual errors damp a signal by F_{ps}. We keep κN of N shots.
- The rescaled estimate has error $\sim \sigma/(F_{\mathrm{ps}}\sqrt{\kappa N})$.
- So R^{-2} is exactly the overhead of **detection followed by rescaling**.
- No checks: $R = F_0$. Gain $G = R/F_0 > 1$ means **the sentinels paid for themselves**.

METHODS

The six challenge steps, all in Qiskit

1 Payloads
6-qubit brick-work Clifford, depth 6, 12, 18; 1 to 16 T
`QuantumCircuit`

2 Reference
state fidelity $\langle\psi|\sigma|\psi\rangle$
`StabilizerState` `Statevector`

3 Checks
one ancilla per target, gamma + windowed
`add_pauli_checks`

4 Noise
announced 1% model and FakeFez calibration
`NoiseModel` `from_backend`

5 Measure
F_0, F_{ps}, κ, cost, broken checks
`Aer density matrix`

6 Compare
all 5,412 moves, best trade-off per level
`SamplerV2` `50k shots`

Exact density matrices remove shot noise, so moves that differ by a fraction of a percent are ranked reliably.

SCALE AND RIGOUR

At a glance

5,412
legal moves, each simulated exactly

$\times$1.13
best fidelity-rate gain (deep, safe *T*)

90 $\rightarrow$ 14
safe *T* sites with 4 checks, depth 18

1.4σ
max deviation vs 50k-shot SamplerV2

5
random payload seeds per experiment

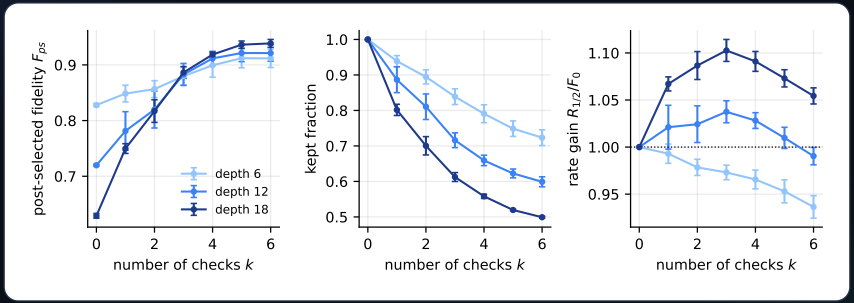
2
noise models: QFF 1% and FakeFez

17
tests in CI, reproducible notebook

12 pp
paper, submitted to arXiv quant-ph

FINDING 1

Checks only pay above a break-even depth



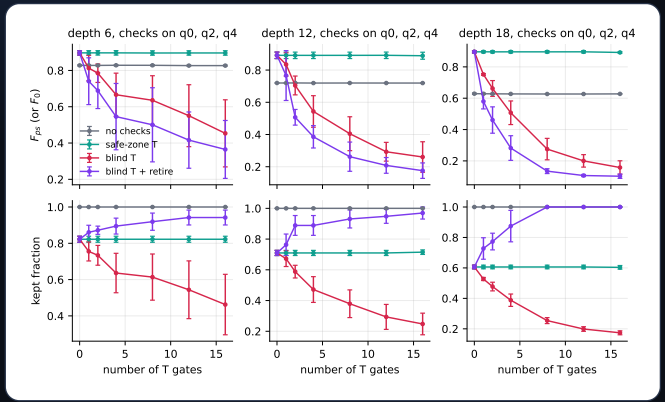
Depth 6: every check lowers R

Depth 12: gain 1.037 at $k = 3$

Depth 18: gain 1.103 at $k = 3$

FINDING 2

Where you put the magic decides everything



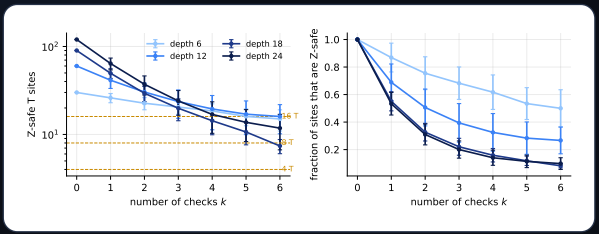
Depth 18, 16 *T* gates, five payloads:



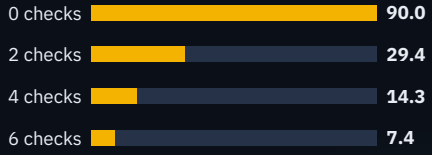
post-selected fidelity F_{ps}

FINDING 3

Protection and magic compete for the same spacetime



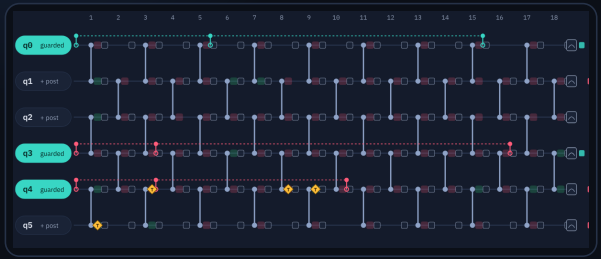
Z-safe *T* sites at depth 18 (of 90):



Every check you add removes places where magic can go. **25 moves** in the game are illegal for exactly this reason.

FINDING 4

A broken check is worse than no check



Blind placement: *T* gates land on X/Y-covered sites and two checks turn red.

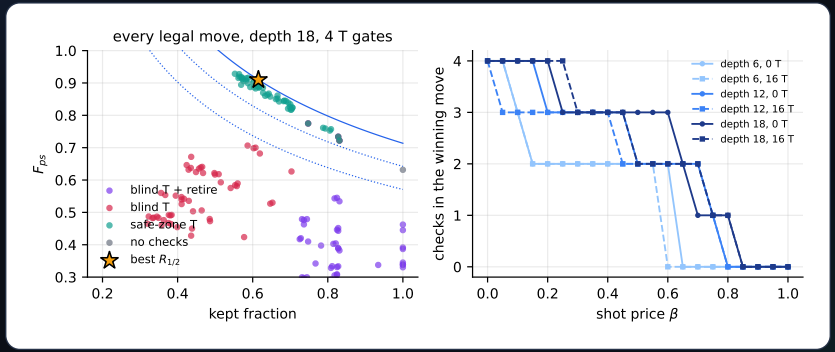
- Blind *T* placement broke **2.4 of 3** checks on average (depth 18, 4 *T*).
- A broken check fires on perfect runs and throws good shots away.
- **Ignoring it does not help:** its gates still entangle the ancilla with a non-Pauli operator.

Depth 18, 4 *T*

blind + retire **0.282** vs no checks **0.627**

TRADE-OFF

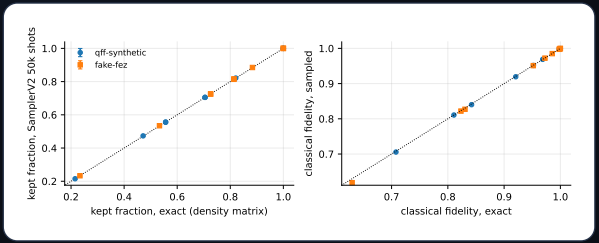
The best move, and why it wins



best move: q0,q3,q4, safe $F_0 = 0.632 \rightarrow F_{ps} = 0.910$ $\kappa = 0.615$ **gain** $\times 1.130$

VALIDATION

Can you trust the numbers?



Exact vs shots

20 configurations re-run with `SamplerV2`, 50k shots: max deviation **1.4**σ.

Two references agree

Stabilizer-group identity equals statevector overlap to machine precision.

Sanity

Noiseless runs: valid checks never fire.

STACK

Qiskit inside

qiskit-paulice 0.2.1

`add_pauli_checks`
`NoiseModel`
`from_backend(FakeFez)`
`uncovered_paulis`
`get_postselection_method`

Qiskit 2.5 core

`QuantumCircuit`
`StabilizerState`
`Clifford`
`Statevector`
`hellinger_fidelity`

Qiskit Aer 0.17

density-matrix simulator
`SamplerV2` primitive
GPU switch
fake backends from
`qiskit-ibm-runtime`

open source, Apache 2.0 | CI: tests + lint | one-command rebuild | Slurm array jobs

SCOPE

Honest limitations

What we simplified

- 6 data qubits and ≤ 4 checks, so every move is exact.
- Uniform Pauli noise, no idling noise (as in the check picker).

Why it still holds

- Coherent syndrome parity gives identical statistics to classical XOR.
- The web game replays Qiskit results; `sentinels play` computes live.

Next: larger circuits on GPU nodes, idling noise, and a hardware run on an IBM device to close the loop between the game and the chip.

THE TAKEAWAY

Protection and magic compete for the same spacetime.

Spend your ancillas where the circuit is deep, keep *T* gates in safe zones, and never trust a broken check.

Play the game

sumitchongder.github.io
spacetime-sentinels

Code, data, video

github.com/Sumitchongder
spacetime-sentinels

Read the report

report/main.pdf
arXiv quant-ph (submitted)

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Thank you